

NMAM INSTITUTE OF TECHNOLOGY, NITTE*(An Autonomous Institution affiliated to VTU, Belagavi)***Seventh Semester B.E. (CSE) (Credit System) Degree Examinations**

January 2023

19CS702 - COMPILER DESIGN

Duration: 3 Hours

Max. Marks: 100

Note: Answer Five full questions choosing Two full questions from Unit – I & Unit – II each and One full question from Unit – III.

Unit – I		Marks	BT*	CO*	PO*
1. a)	Explain the concept of Buffer Pairs, input buffering scheme in the lexical analysis. Also write the algorithm for Buffer Pairs scheme.	10	L*2	1	1
b)	Write the regular definition for if statement and also construct the transition diagram for relational operators (<, <=, >, >=, =, !=), identifiers.	10	L3	1	2
2. a)	Explain in detail various Phases of compilers and apply the same for the input string ans = ((a + c) * b / c - 60) + i (assume all the identifiers are of real type).	10	L3	1	2
b)	Design a LEX program. i) To recognize signed numbers ii) Write a LEX program to read 5 student names from keyboard and display longest name entered.	10	L3	1	1
3. a)	Define Left recursion and Left factoring and apply the same for the grammar E → E * T T T → id + T id	5	L3	2	2
b)	Construct Predictive Parsing table for the following Grammar and check whether grammar is LL(1) or not. S → (L) ε L → L, S S	10	L3	2	2
c)	What are predictive parser? Consider the grammar given below. S → aAb A → cd c Construct the parse tree for the input string "acb" using recursive descent parser (with backtracking).	5	L3	2	2
Unit – II					
4. a)	Write an algorithm for constructing LR(1) items.	5	L2	2	2
b)	Construct SLR(1) parsing table for the following grammar E → BB B → c B d	10	L3	3	2
c)	Explain Quadraple, triple and indirect triples. Translate the following expression, b*-c+(a+c) into Quadraple, triple and indirect triples.	5	L3	3	2
5. a)	Consider the following grammar: S → aAd bBd aBe bAe A → c B → c i) Construct sets of LR(1) items ii) Construct canonical LR(1) parsing table iii) Show the parsing steps for the string "baaba"	10	L3	3	2
b)	Define the following with examples: i) S-attributed definitions ii) L-attributed definitions	4	L2	4	2

P.T.O.

- c) With suitable logical diagrams, build a SDD to generate 3 address code for the following flow control statements:
 $S \rightarrow \text{if } E \text{ then } S_1 \text{ else } S_2 \mid \text{while } E \text{ then } S_1.$ 6 L3 4 2
6. a) Compare and contrast SLR with LALR. Show the following grammar is LALR(1)
 $S \rightarrow Aa \mid bAc \mid dc \mid bda$
 $A \rightarrow d$ 10 L3 3 2
- b) For the grammar
 $L \rightarrow En$
 $E \rightarrow E_1 + T$
 $E \rightarrow T$
 $T \rightarrow T_1 * F$
 $T \rightarrow F$
 $F \rightarrow (E)$
 $F \rightarrow \text{digit}$
 i) Obtain the SDD for the above
 ii) Construct annotated parse tree for the string $3*5+4n$ 10 L3 4 2
- Unit – III**
7. a) Discuss the issues in the design of a code generator. 10 L2 5 2
- b) What is a Basic block? Write an algorithm to partition the three address instructions into basic blocks. Construct DAG for the following code block.
- i) $T_1 = 4*i$
 - ii) $T_2 = a[T_1]$
 - iii) $T_3 = 4*i$
 - iv) $T_4 = b[T_3]$
 - v) $T_5 = T_2 * T_4$
 - vi) $T_6 = \text{prod} + T_5$
 - vii) $\text{Prod} = T_6$
 - viii) $T_7 = i + 1$
 - ix) $i = T_7$
 - x) if $i \leq 40$ goto (i)
- 10 L3 5 2
8. a) Define flow graph. Explain how it is constructed. 4 L2 5 2
- b) What is the need for optimization? List and explain any two local optimization methods. 6 L2 5 2
- c) Generate the 3-address statement for the following programming construct and obtain the basic blocks for the generated code.
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i=1
do
 sum = sum + a[i]*b[i]
 i=i+1
while(i<=20)

```
- 10 L3 5 2